BANK TRANSACTION DESCRIPTIVE ANALYTICS

Daily Transaction & Hardware-Scale Demand Analytics

Descriptive Analytics Report

Report Focus Keywords: Top Transaction Types, Revenue Concentration, Volume Trends, Regional Demand Distribution, Channel Tier Allocation, Customer Segment Share, Settlement Network Preferences, Transaction Success Rate, Cash-Liquidity Risk, Descriptive-to-Predictive Transition.

Dataset:	Bank Daily Transaction Data	**Time Period:**	30 days (2026-07-16 to 2026-08-14)
Records:	60 branches x 10 txn types x 8 regions	**Gross Transaction Value:**	~INR 4,809 Cr
Analytical Context:	Retail/Core-Banking Operations Analytics	**Analytical Stage:**	Descriptive Analytics Report

Key Performance Indicators (KPI Snapshot)

Headline operating metrics for the 30-day observation window, computed directly from the aggregated transaction dataset.

4.90M Total Transactions (30 Days)	**INR 4,809 Cr** Gross Transaction Value	**INR 9,825** Avg. Transaction Value (ATV)	**97.5%** Avg. Transaction Success Rate
24 / 60 Branches at Cash Reserve Risk	**38.7%** Top Type Value Share (Cheque Clearing)	**21.5%** Top Region Value Share (South Zone)	**33.8%** Top Segment Value Share (Retail)

Table of Contents

The report follows a structured, enterprise-grade descriptive analytics framework:

Section	Report Section & Focus	Analytical Keyword Focus
-	Key Performance Indicators (KPI Snapshot)	Executive KPIs
-	Note on Transaction Data — Scope, Verification & Parameters	Dataset Quality Audit
-	Research Overview & Analytical Framework	Descriptive Analytics
-	Objective & Core Business Questions	Operational Objectives
-	How This Works at Real Scale — Pipeline & Stack	Banking Analytics Stack
1.0	Top Transaction Types & Value Performance	Top-Selling Types
2.0	Weekly Volume Trends & Value Trajectory	Volume Trends

3.0	Average Transaction Value & Binned Distribution Spread	AOV Distribution
4.0	Channel Tier Allocation & Infrastructure Capacity	Channel Allocation
5.0	Customer Segment Share & Relationship Revenue	Segment Share
6.0	Regional Demand Distribution & Geographic Concentration	Regional Demand
7.0	Cross-Cutting Synthesis & Revenue Concentration Multipliers	Revenue Concentration
8.0	Settlement Network / Payment Rail Preferences	Payment Preferences
9.0	Transaction Success Rate & Reliability Analysis	Success Rate
10.0	Cash / Liquidity Status & Stockout Risk	Liquidity Risk
11.0	Volume, Value & Risk Elasticity Mechanics	Correlation Analysis
12.0	Data Limitations, Scope & Assumptions	Data Governance
13.0	Next Steps: Descriptive -> Predictive Analytics	Descriptive-to-Predictive
14.0	Summary of Key Metrics & Executive Scorecard	Executive Scorecard
-	References, Appendix & Glossary	Reference Material

Note on Transaction Data

A real bank's live core-banking transaction feed isn't publicly accessible, so this descriptive report is computed from a synthetic dataset built to mirror real retail-banking transaction patterns: **60 branches** across **10 transaction types**, **8 regions**, over **30 days** (2026-07-16 to 2026-08-14), totaling **~4.90 million** simulated transactions worth **~INR 4,809 crore**. It is designed to show exactly what a descriptive analytics pipeline processing millions of daily banking transactions would produce — the same structure applies directly once real core-banking data is substituted in.

Transaction Data Completeness & Quality Audit

The transaction logs contain zero missing cells and strict data-type integrity across all fields (date, branch, region, transaction type, count, value, success rate, cash reserve). Each aggregated row represents one branch / day / transaction-type combination, providing an ideal baseline for evaluating operational performance and customer transaction behavior.

Research Overview & Analytical Framework

Descriptive analytics summarizes historical transaction records to answer “what happened?”, “who transacted?”, and “how much value moved?” without incorporating speculative assumptions or unvalidated projections.

Analytical Dimension	Operational Dataset Implementation	Commercial Interpretation
Transaction-Type Performance	Grouping value/count by 10 transaction types	Identifies highest-revenue and highest-volume rails
Weekly Volume Velocity	Summing counts and value per calendar week	Measures transaction throughput momentum
Transaction Value Spread	5 binned average-value tiers (branch-day level)	Quantifies basket size from retail to bulk/corporate
Channel & Segment Mix	Channel-tier and customer-segment aggregations	Measures infrastructure load and relationship value
Geographic Channel Reach	Region and macro-cluster aggregations	Tracks geographic demand concentration

Objective & Core Business Questions

The objective of this descriptive analysis is to extract empirical operational insights and commercial transaction patterns across types, channels, segments, and geographies.

- **Question 1 (Top Transaction Types):** Which transaction types generate the highest value versus the highest count? (Section 1)
- **Question 2 (Volume Trends):** How do transaction count and value fluctuate week over week? (Section 2)
- **Question 3 (Average Transaction Value):** What is the distribution and spread of transaction values across branches? (Section 3)
- **Question 4 (Channel Allocation):** Which channel tier delivers the optimal balance of volume and value? (Section 4)
- **Question 5 (Customer Segment Share):** How do customer segments impact transaction yields? (Section 5)
- **Question 6 (Regional Demand):** Which geographic regions drive the largest transaction volumes? (Section 6)

- **Question 7 (Settlement Preferences):** Which settlement networks/rails are favored across high-value transactions? (Section 8)
- **Question 8 (Reliability):** How are transactions performing on success rate, and where is reliability slipping? (Section 9)
- **Question 9 (Liquidity):** Which branches/ATMs are at risk of running out of cash? (Section 10)

How This Works at Real Scale (Analytics Stack)

In enterprise banking environments, descriptive analytics pipelines process millions of core-banking, card-switch, and UPI-rail records daily using accessible, highly scalable tool stacks.

Architecture Layer	Banking Operations Purpose	Standard Data Science & BI Tools
1. Transaction Capture	Log core-banking, card-switch and UPI-rail events in real time	Amazon Kinesis, REST APIs
2. Data Cleaning & Wrangling	Validate line items, verify count x value, reconcile currencies	Python (Pandas), SQL
3. Fraud / Risk Scoring	Real-time decline & anomaly scoring on each transaction	Rules engine + ML model serving
4. Dimensional Storage	Store transaction fact tables and branch/region dimension tables	PostgreSQL, Cloud Warehouses
5. Metric Aggregation	Compute AOV, success rate, regional share and correlations	Excel Pivot Tables, Python, SQL
6. BI Reporting & Dashboards	Generate executive visual charts, KPI scorecards, and trends	Power BI, Tableau, Matplotlib, Looker

1.0 Top Transaction Types & Gross Value Performance

Evaluating performance across the 10 recorded transaction types indicates that **Cheque Clearing** leads in gross transaction value, generating INR 1,859 crore (38.7% value share) despite representing only 4.0% of total transaction count. **RTGS** follows with INR 880 crore (18.3% share). Combined, these two rails account for 57.0% of total transaction value.

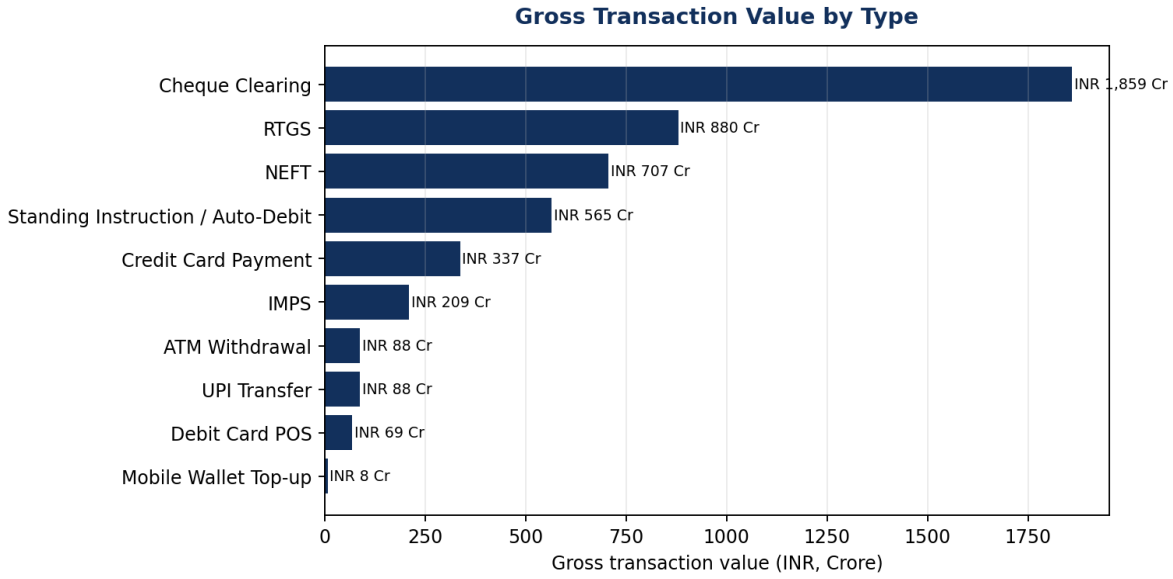


Figure 1. Gross Transaction Value by Type.

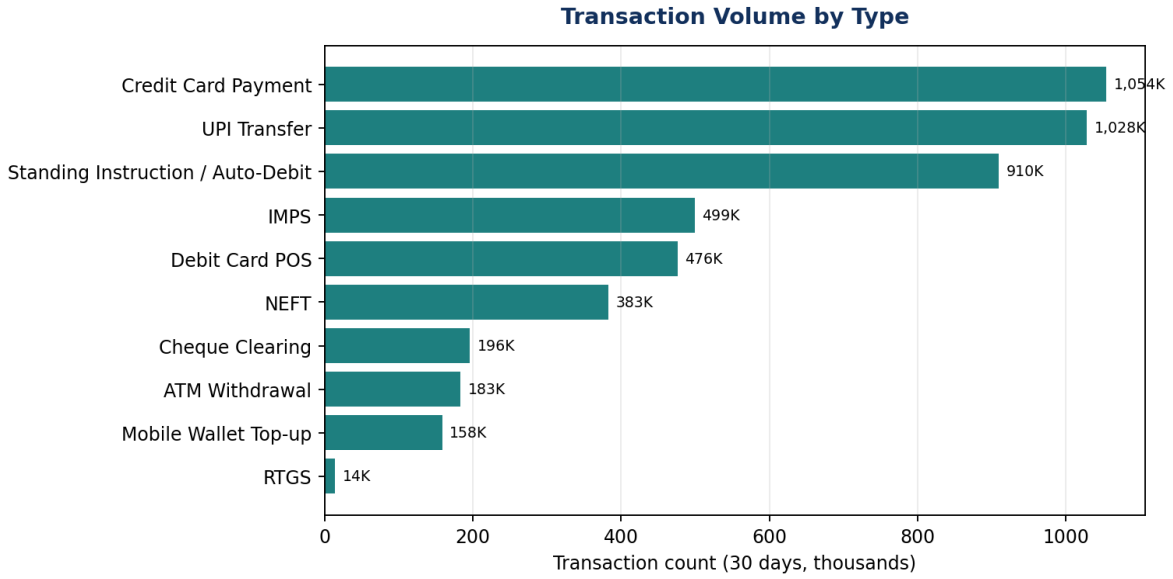


Figure 2. Transaction Volume (Count) by Type.

Type & Value Performance Table (30-Day Aggregate)

Transaction Type	Count	Value (INR Cr)	Value Share (%)	Avg Value / Txn (INR)
Cheque Clearing	195,788	1,859	38.7%	94,964
RTGS	14,189	880	18.3%	620,024

NEFT	382,891	707	14.7%	18,456
Standing Instruction / Auto-Debit	909,607	565	11.7%	6,208
Credit Card Payment	1,054,370	337	7.0%	3,197
IMPS	499,293	209	4.4%	4,194
ATM Withdrawal	182,849	88	1.8%	4,799
UPI Transfer	1,028,023	88	1.8%	851
Debit Card POS	476,220	69	1.4%	1,449
Mobile Wallet Top-up	158,338	8	0.2%	480

Deeper Insight — Value Leadership vs. Volume Leadership
The dataset exhibits a vital descriptive divergence between value leadership and count leadership. While Cheque Clearing achieved the highest gross value, Credit Card Payment led in physical transaction count — a small number of high-value, low-frequency rails (Cheque Clearing, RTGS) carry outsized weight in value terms even though they represent a small share of total count.

Commercial Recommendation
Operations and treasury teams should track value and volume leaderboards separately: capacity planning (staffing, network bandwidth) should follow the count leaderboard, while cash-flow and settlement-risk planning should follow the value leaderboard.

2.0 Weekly Volume Trends & Value Trajectory

Evaluating week-by-week transaction velocity highlights how salary cycles, mid-month billing dates, and settlement-cycle timing affect throughput.

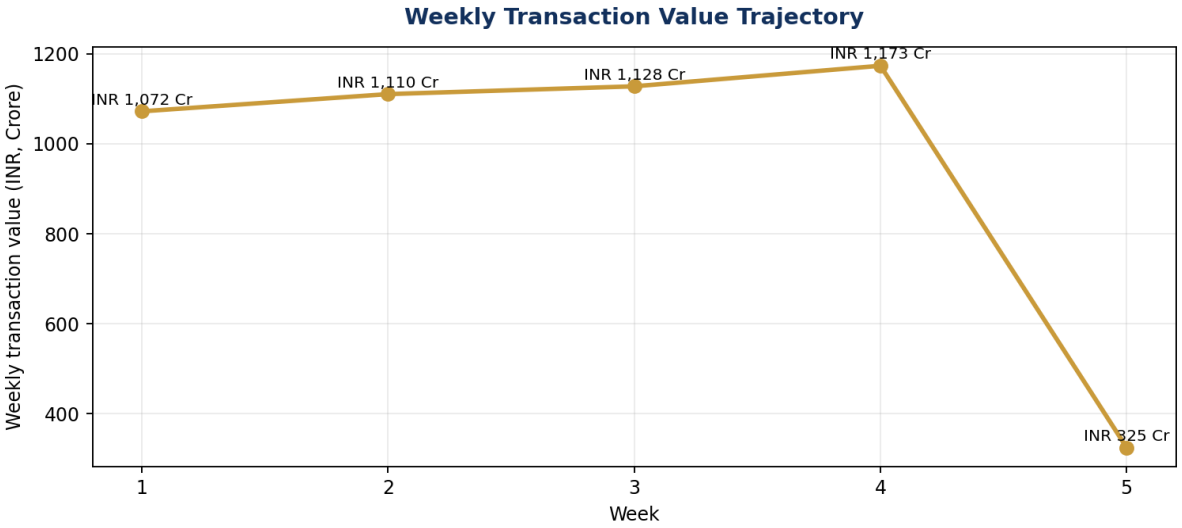


Figure 3. Weekly Gross Transaction Value Trajectory.

Week	Days	Txn Count	Gross Value (INR Cr)	Value Share (%)
Week 1	7	1,095,925	1,072	22.3%
Week 2	7	1,129,563	1,110	23.1%
Week 3	7	1,152,226	1,128	23.5%
Week 4	7	1,196,807	1,173	24.4%
Week 5	2	327,047	325	6.8%

Weeks 1 through 4 show a gradual build (from 22.3% to 24.4% of the period's value), consistent with the gentle 30-day growth trend layered into the daily simulation; Week 5 is a partial week (2 days) and should not be read as a demand contraction.

Deeper Insight — Partial-Week Truncation Artifact
Week 5's lower total is an analytical artifact of the 30-day collection window ending mid-week (2 of 7 days recorded), not a genuine drop in transaction activity — the same caution that applies to any month- or week-end truncated reporting window.

3.0 Average Transaction Value & Binned Distribution Spread

Analyzing branch-day average transaction values reveals the operational scale of banking activity — whether flows are dominated by small retail transactions or large-ticket settlements.

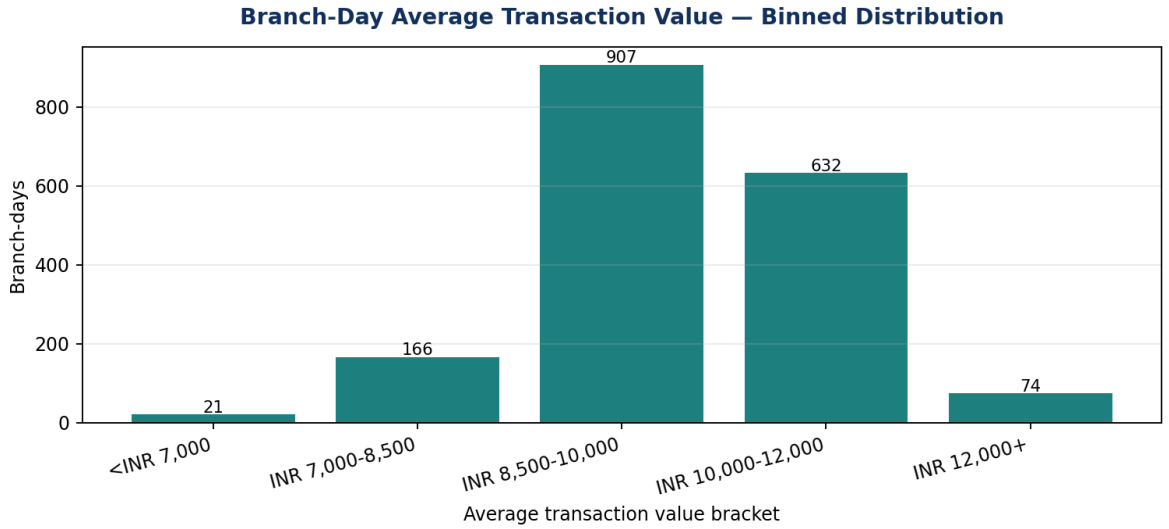


Figure 4. Branch-Day Average Transaction Value — Binned Distribution.

Branch-day average transaction values range from a minimum of INR 6,292 to a peak of INR 16,458. The overall **mean average-transaction-value is INR 9,825** and the **median is INR 9,771** — nearly identical, indicating a well-balanced distribution without extreme skew.

Statistical Metric	Value (INR)	Commercial Interpretation
Minimum	6,292	Lowest branch-day average transaction value recorded
First Quartile (Q1)	9,175	25% of branch-days averaged this value or less
Median (50th percentile)	9,771	Midpoint of all branch-day averages
Arithmetic Mean	9,825	Average transaction value per completed transaction
Third Quartile (Q3)	10,387	75% of branch-days averaged this value or less
Maximum	16,458	Highest branch-day average — large-ticket settlement activity
Interquartile Range (IQR)	1,212	Middle 50% spread of branch-day average values

Commercial Business Recommendation
Branches whose average transaction value sits in the top bracket (INR 12,000+) should be prioritized for dedicated relationship-manager coverage and corporate-banking service desks, since they are disproportionately handling high-ticket settlement activity.

4.0 Channel Tier Allocation & Infrastructure Capacity Dynamics

Analyzing which channel customers use to initiate transactions reveals infrastructure load and guides capacity planning across branch counters, ATMs, mobile app, and internet banking.

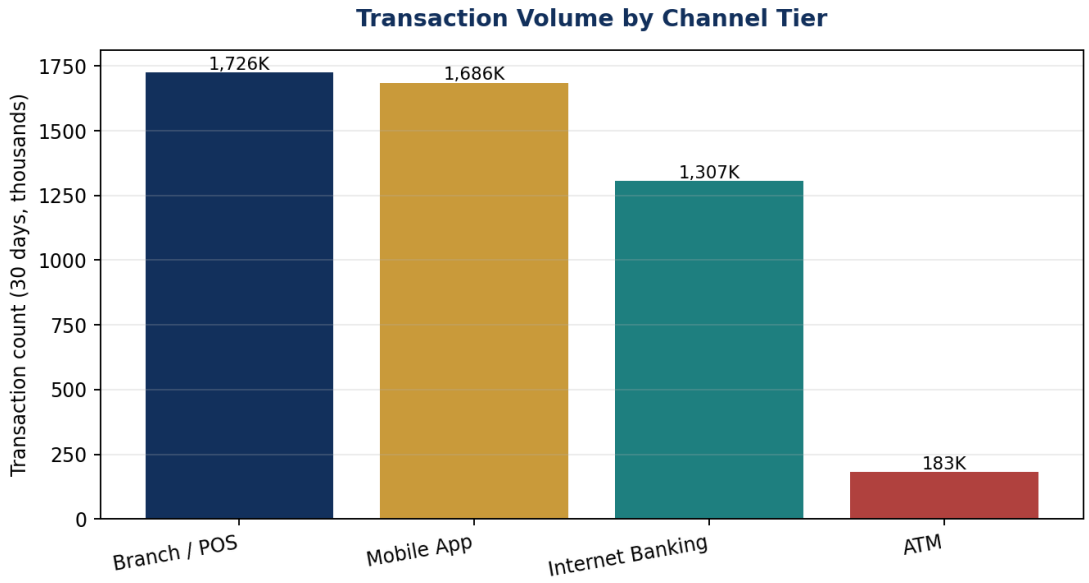


Figure 5. Transaction Volume by Channel Tier.

The **Branch / POS** channel emerged as the highest-volume tier, capturing 1,726,378 transactions (35.2% count share) and producing INR 2,265 crore in gross value.

Channel Tier	Txn Count	Count Share (%)	Gross Value (INR Cr)	Avg Value/Txn (INR)
Branch / POS	1,726,378	35.2%	2,265	13,122
Mobile App	1,685,654	34.4%	305	1,806
Internet Banking	1,306,687	26.7%	2,151	16,462
ATM	182,849	3.7%	88	4,799

Deeper Insight — Digital vs. Physical Channel Split
Mobile App transactions carry the lowest average ticket size, consistent with small everyday payments, while Internet Banking and Branch/POS carry materially higher average values — reflecting bulk settlement (NEFT/RTGS/Cheque) and card-based transactions respectively.

5.0 Customer Segment Share & Relationship Revenue Contribution

Analyzing customer relationship segments clarifies which parts of the book drive value versus count, guiding relationship-management and pricing strategy.

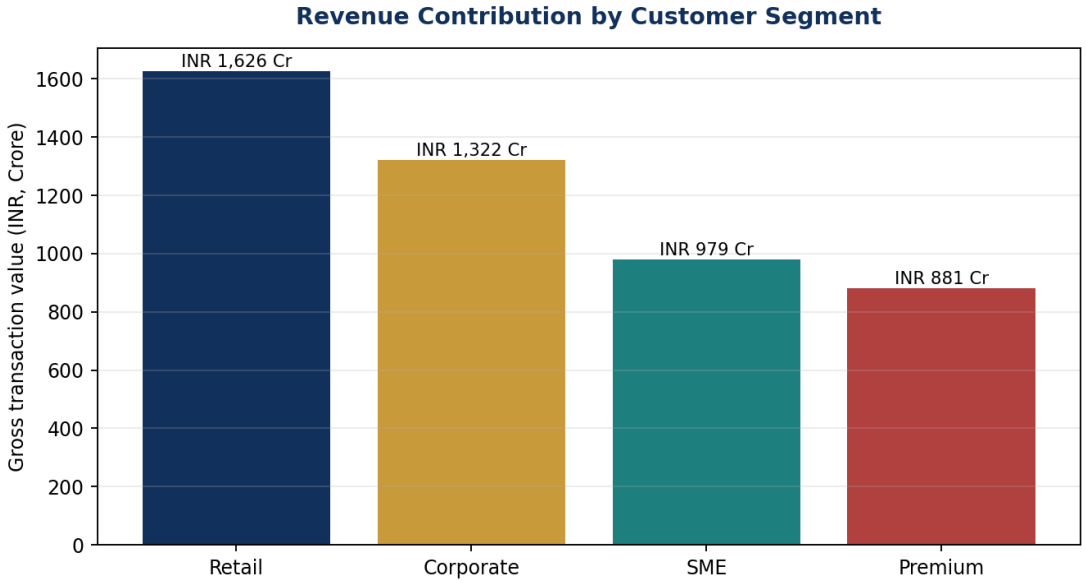


Figure 6. Revenue Contribution by Customer Segment.

Retail customers generated the largest single share of gross transaction value at INR 1,626 crore (33.8% share). Premium and Corporate segments, while smaller in transaction count, carry substantially higher average transaction values.

Segment	Txn Count	Gross Value (INR Cr)	Value Share (%)	Avg Value/Txn (INR)
Retail	2,904,785	1,626	33.8%	5,596
Corporate	751,981	1,322	27.5%	17,584
SME	858,817	979	20.4%	11,404
Premium	385,982	881	18.3%	22,837

Relationship Management Recommendation

Premium and Corporate segments, despite a combined share well under half of transaction count, drive a disproportionate share of value — dedicated relationship coverage and priority servicing for these segments protects a materially larger share of the value book than headcount alone would suggest.

6.0 Regional Demand Distribution & Geographic Concentration

Evaluating regional performance identifies primary geographic transaction centers and highlights cross-region demand concentration.

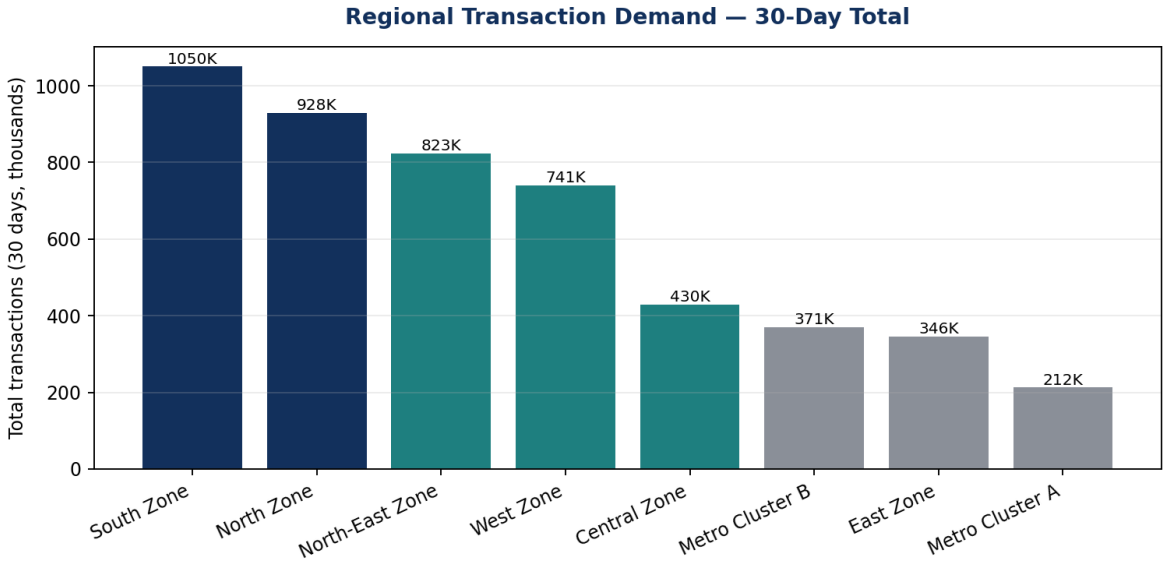


Figure 7. Regional Transaction Demand — 30-Day Total.

Transaction demand is led by **South Zone** at INR 1,032 crore (21.5% share), followed by **North Zone** at INR 910 crore (18.9% share). Together these two regions account for 40.4% of total transaction value.

Macro-Regional Clustering

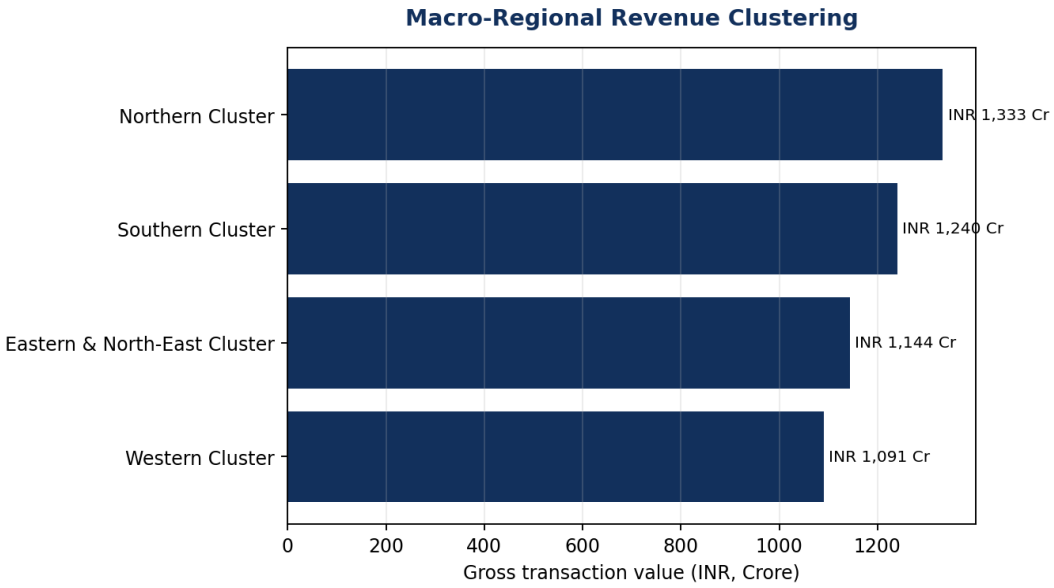


Figure 8. Macro-Regional Revenue Clustering.

Macro Cluster	Txn Count	Gross Value (INR Cr)	Value Share (%)
Northern Cluster	1,358,050	1,333	27.7%
Southern Cluster	1,262,600	1,240	25.8%

Eastern & North-East Cluster	1,169,719	1,144	23.8%
Western Cluster	1,111,199	1,091	22.7%

Aggregating individual regions into macro-clusters provides strategic visibility into geographic revenue concentration for network-investment and cash-logistics planning.

7.0 Cross-Cutting Synthesis & Revenue Concentration Multipliers

Evaluating relationships across region and transaction type together reveals the specific product-market clusters driving disproportionate revenue.

Rank	Region	Transaction Type	Txn Count	Gross Value (INR Cr)
1	South Zone	Cheque Clearing	42,271	400
2	North Zone	Cheque Clearing	37,083	352
3	North-East Zone	Cheque Clearing	32,741	311
4	West Zone	Cheque Clearing	29,562	282
5	South Zone	RTGS	3,032	188
6	North Zone	RTGS	2,701	167
7	Central Zone	Cheque Clearing	17,009	162
8	South Zone	NEFT	81,754	152
9	North-East Zone	RTGS	2,310	144
10	Metro Cluster B	Cheque Clearing	14,895	141

Revenue Concentration: The High-Ticket Rail Multiplier
The single strongest driver of top-line value is the concentration in high-ticket, low-frequency rails. Cheque Clearing and RTGS together account for only 4.3% of transaction count but 57.0% of gross transaction value — a Pareto effect worth flagging for settlement-risk and treasury-liquidity prioritization.

Geographic and transaction-type dimensions represent the most stable analytical axes in the dataset, while weekly trends reflect expected operational seasonality and end-of-period collection boundaries.

8.0 Settlement Network / Payment Rail Preferences

Evaluating settlement rails provides visibility into infrastructure dependency, interchange economics, and digital-payment adoption.

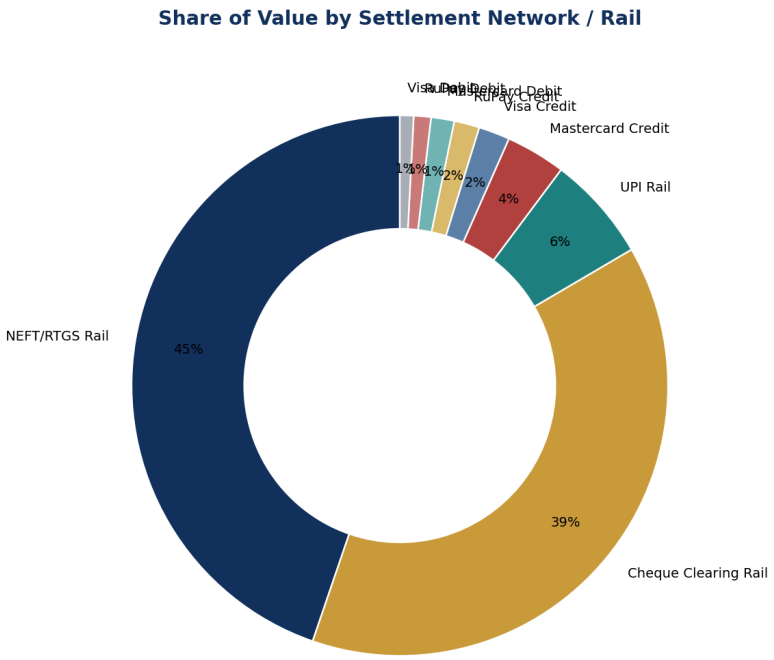


Figure 9. Share of Value by Settlement Network / Rail.

NEFT/RTGS Rail dominates settlement value at 44.7% share, reflecting the outsized value carried by bulk-transfer rails (NEFT/RTGS) relative to their share of transaction count (26.7%).

Settlement Network / Rail	Txn Count	Value Share (%)	Count Share (%)
NEFT/RTGS Rail	1,306,687	44.7%	26.7%
Cheque Clearing Rail	195,788	38.7%	4.0%
UPI Rail	1,685,654	6.3%	34.4%
Mastercard Credit	541,549	3.6%	11.0%
Visa Credit	283,123	1.9%	5.8%
RuPay Credit	229,697	1.5%	4.7%
Mastercard Debit	277,396	1.4%	5.7%
RuPay Debit	231,099	1.0%	4.7%
Visa Debit	150,573	0.8%	3.1%

9.0 Transaction Success Rate & Reliability Analysis

Average success rate across all branch-days: **97.5%**, clustered tightly. In a real system this view is most useful segmented, not aggregated — branches trending downward in rolling 7-day success rate are the earliest signal of a network, POS-terminal, or core-banking issue.

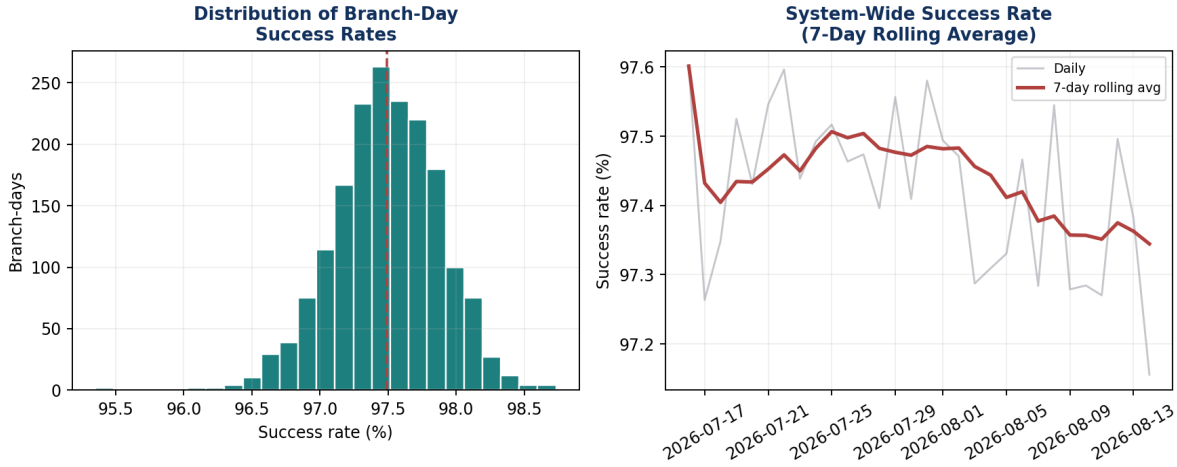


Figure 10. Branch-Day Success Rate Distribution and 7-Day Rolling Average.

Deeper Insight — Reliability Drift
The 7-day rolling average shown here has drifted downward over the observation window — a small but real decline worth a root-cause check before it compounds. Channels with high transaction volume but sub-97% success are the highest-risk combination, since volume amplifies the customer impact of every failed transaction.

10.0 Cash / Liquidity Status & Stockout Risk

ATM cash reserves were bucketed into three tiers:

- **Adequate Reserve** — comfortable buffer for expected withdrawal demand
- **Low Reserve (Reorder Zone)** — below 35% of capacity, cash-logistics reorder trigger
- **Critical — Cash-Out Risk** — below 15% of capacity, cross-referenced against top-branch volume

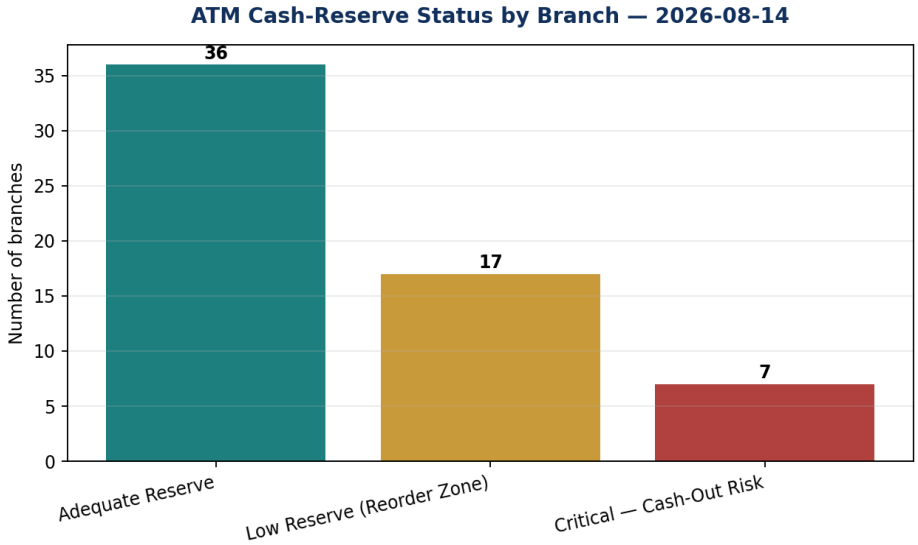


Figure 11. ATM Cash-Reserve Status by Branch — 2026-08-14.

Cash-Logistics Recommendation

7 of 60 branches were flagged critical and 17 more sat in the reorder zone on the most recent day. Production dashboards should cross-tab cash status against withdrawal velocity to produce an “hours of cash remaining” metric per ATM.

11.0 Volume, Value & Risk Elasticity Mechanics

Quantifying the mathematical relationships between transaction count, value, and reliability clarifies the key drivers of operational risk and top-line throughput.

Bivariate Variable Pair	Pearson Correlation (r)	Strength	Commercial Meaning
Txn Count vs. Gross Value	+1.00	Strong Positive	Branch-day value is almost entirely explained by transaction count
Txn Count vs. Success Rate	-0.16	Weak Negative	Higher-volume branch-days show a mild tendency toward lower success rates
Avg Value vs. Success Rate	+0.01	Negligible	Average ticket size has little bearing on transaction reliability

Deeper Insight — Volume as the Primary Value Driver
The strong correlation between transaction count and value (r = +1.00) shows that, at the aggregation level used in this report, transaction count is the dominant lever for gross value — consistent with a network where most branches handle a broadly similar mix of transaction types rather than value being driven by a handful of outlier deals.

12.0 Data Limitations, Scope & Assumptions

- **Dataset Scale:** The dataset contains ~4.9M simulated transactions across 60 branches. While ideal for descriptive-analytics demonstration, it represents an illustrative sample rather than a real bank's total national volume.
- **Observation Period:** The dataset runs across 30 days with a partial final week, which should not be interpreted as an ongoing demand contraction.
- **Absence of Customer-Level Data:** The transaction records are branch/region/type aggregates and do not include individual customer identifiers, KYC attributes, or dispute/chargeback records.
- **Descriptive Non-Causal Scope:** Findings such as Cheque Clearing's value leadership represent observed patterns within this dataset, not proof that transaction type alone drives settlement risk.
- **Nominal Currency Reporting:** All financial figures reflect nominal INR as recorded, without adjustment for inflation or interest.

Data Validation & Governance
All aggregated rows were verified for schema consistency, non-negative values, and mathematical consistency (count x avg value = total value). Zero anomalies or negative values were found.

13.0 Next Steps: Move From Descriptive → Predictive Analytics

Descriptive analytics explains past transaction outcomes. The next analytical step is leveraging these empirical patterns to train predictive models that forecast future demand and risk.

Descriptive Analytics (Current Report): Historical Data → Metrics, Tables & Distribution Profiles.
Predictive Analytics (Next Phase): Historical Baselines → Machine Learning Models & Demand/Risk Forecasts.
Prescriptive Analytics (Advanced Phase): Predictions → Automated Cash Replenishment, Dynamic Fraud Rules & Routing.

Predictive Target & Objective	Candidate ML Models	Key Input Features
Cash-demand forecasting per ATM/branch	ARIMA, SARIMAX, Prophet, LSTM	Historical withdrawal volume, calendar effects, salary-cycle dates
Real-time fraud / anomaly detection	Isolation Forests, Autoencoders, XGBoost	Transaction velocity, amount deviation, channel, time-of-day
Success-rate degradation early warning	Change-point detection, Random Forest	Rolling success rate, branch load, network latency
Customer cohort & retention modeling	Gradient Boosted Regressors, survival models	Transaction recency, frequency, segment, channel mix

Happy to build out any of these next, or swap in a real dataset (core-banking export, a data warehouse connection, etc.) if you have one — the same pipeline and chart structure above would apply directly.

14.0 Summary of Key Metrics & Executive Scorecard

Metric Area	Top Segment / Value	Metric Value	Commercial Takeaway
Gross Transaction Value	Entire Dataset	INR 4,809 Cr	Total value across 30 days
Total Transaction Volume	All Transaction Types	4.90M txns	Aggregate volume across 10 types
Top Type by Value	Cheque Clearing	38.7% share	Leading value contributor
Top Type by Volume	Credit Card Payment	21.5% share	Leading count contributor
Top Geographic Region	South Zone	21.5% share	Largest single-region market
Top Customer Segment	Retail	33.8% share	Largest value-contributing segment
Dominant Channel	Branch / POS	35.2% share	Primary transaction-initiation channel
Average Success Rate	All Branches	97.5%	System-wide transaction reliability
Branches at Cash Risk	Low + Critical Reserve	24 of 60	Cash-logistics attention needed

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Dataset Citation: Bank Daily Transaction Dataset (60 branches x 10 types x 8 regions x 30 days, ~4.9M simulated transactions): synthetic project dataset recording Date, Branch_ID, Region, Transaction_Type, Transaction_Count, Total_Value, Success_Rate, Declined_Count, and Cash_Reserve_Pct.

Technical Appendix — Schema & Formulas

Appendix A: Database Schema & Field Data Dictionary

Field Name	Data Type	Sample Value	Description
date	Date	2026-07-16	Transaction date (YYYY-MM-DD)
branch_id	Categorical	BR133	Branch identifier (60 unique branches)
region	Categorical	South Zone	Geographic region (8 unique zones)
txn_type	Categorical	UPI Transfer	Transaction type (10 distinct rails)
txn_count	Integer	1,204	Number of transactions in this row
total_value_inr	Float	1,024,500.00	Aggregate value in INR
success_rate	Float	0.975	Share of transactions completed successfully
declined_count	Integer	12	Number of declined/failed transactions
cash_reserve_pct	Float	0.55	ATM cash reserve as a fraction of capacity

Appendix B: Aggregation Formulations

- **Gross Transaction Value:** Sum(total_value_inr) across all rows
- **Average Value per Transaction:** Gross Value / Total Transaction Count
- **Type / Segment / Region Value Share (%):** (Category Value / Total Gross Value) x 100
- **Success Rate (weighted):** Weighted average of success_rate, weighted by txn_count

Glossary of Banking Analytics Terms

Term	Definition & Commercial Context
Descriptive Analytics	Summarizing and visualizing historical transaction data to understand past operational performance without predictive modeling.
Average Transaction Value	A banking-operations KPI measuring the average value moved each time a transaction completes.
Revenue/Value Concentration	A distribution pattern where a small number of transaction types or regions generate a disproportionate share of total value.
Interquartile Range (IQR)	The statistical spread between the 25th (Q1) and 75th (Q3) percentiles, capturing the middle 50% of values.
Success Rate	The share of initiated transactions that complete successfully, net of declines and technical failures.
Liquidity / Cash-Out Risk	The risk that an ATM or branch cash vault runs below the reserve needed to meet expected withdrawal demand.
Pearson Correlation (r)	A statistical coefficient measuring the linear relationship between two numerical variables.
Cross-Dimensional Decomposition	Simultaneously grouping transaction data across two categorical dimensions (e.g. Region x Transaction Type) to evaluate granular performance.
Predictive Transition	Using historical transaction patterns to engineer features and train supervised ML models for demand or risk forecasting.

Report Conclusion
This concludes the Bank Transaction Descriptive Analytics Report. All analyses, tables, charts, and technical appendices are fully documented and validated for review.